Huawei Access Network Node MALL-ACC-03 Hardware/Environmental Fault Resolution

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8455

Description

In the Huawei Access Network, Node MALL-ACC-03 is equipped with Advanced Switching ASICs (Application-Specific Integrated Circuits) in its line cards, which are essential for high-speed data transmission and processing. The ASICs employ a complex combination of hardware and software to manage the data flow, including error correction mechanisms to ensure reliable data transmission. However, in rare instances, the ASIC memory parity errors or line card crashes can occur due to various environmental or hardware faults, such as overheating, power supply redundancy loss, or manufacturing defects. These faults can cause the ALARM_8455 to be triggered, indicating a warning-level issue. To resolve this issue, it is crucial to identify and address the root cause of the problem, which may involve troubleshooting the hardware, environmental conditions, or firmware configurations. The following Standard Operating Procedure (SOP) outlines the steps to be taken to diagnose and resolve the Hardware/Environmental Fault on Node MALL-ACC-03.

Pre-requisite Checks

- check: Verify the site location and ensure that the Malleswaram Node is accessible for remote hands or on-site personnel.

- check: Confirm that the redundant link status is operational and not affected by the fault.

- check: Review the system logs to identify any relevant error messages or patterns preceding the ALARM_8455.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to overheating or power supply issues.	High
Line card crashes due to manufacturing defects or firmware misconfigurations.	Medium
Environmental factors such as dust, humidity, or temperature fluctuations affecting the hardware.	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-asic, Expected Output: ASIC memory parity error detected. Line card crash occurred., Description: This command displays the current ASIC status, including any error messages related to memory parity errors or line card crashes.

Command: display environment-monitor, Expected Output: Temperature: 50°C. Humidity: 60%., Description: This command displays the current environmental monitoring data, including temperature, humidity, and other relevant parameters.
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Command: display power-supply, Expected Output: Redundant power supply lost., Description: This command displays the current power supply status, including any issues with redundant power supply loss.
```

Resolution Steps

Step 1: Step: Verify the environmental conditions and ensure that the node is operating within the specified temperature and humidity ranges., Command: display environment-monitor, Description: Verify the current environmental monitoring data to ensure that the node is operating within the specified temperature and humidity ranges.

Step 2: Step: Perform a hard reset of the line card to clear any configuration-related issues., Command: reset linecard {slot}, Description: Perform a hard reset of the line card to clear any configuration-related issues that may be contributing to the fault.

Step 3: Step: Replace the faulty ASIC or line card with a new one., Command: install asic {slot} {asic-type}, Description: Replace the faulty ASIC or line card with a new one to ensure that the node can operate reliably.

Post-Check Verification

```
Command: display current-asic, Expected Output: ASIC memory parity error cleared. Line card operational., Description: Verify that the ASIC memory parity error has been cleared and the line card is operational.

Command: display environment-monitor, Expected Output: Temperature: 45°C. Humidity: 55%., Description: Verify that the environmental conditions are within the specified ranges and not affecting the node's operation.

Command: display power-supply, Expected Output: Redundant power supply restored., Description: Verify that the redundant power supply has been restored and the node is operating reliably.
```

Roll-back Procedures

- Step: Restore the original configuration and settings., Command: undo install asic {slot} {asic-type}, Description: Restore the original configuration and settings to ensure that the node can operate as it did before the resolution steps were taken.

- Step: Reboot the node to clear any residual issues., Command: reboot, Description: Reboot the node to clear any residual issues that may have arisen during the resolution process.